Non-Equilibrium Extraction of Vacuum Zero-Point Energy via Topological Geometric Pumping in an 8-Site Cyclic Quantum Meta-Material

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We report a novel mechanism for steady-state energy extraction from vacuum zero-point fluctuations. By constructing an 8-site cyclic quantum circuit with topological geometric phases, we demonstrate that a dynamic rotation of the boundary conditions leads to the generation of real photons via the dynamical Casimir effect. Incorporating the Lindblad master equation and Kerr nonlinearities, we show that the system reaches a stable negative-energy steady state. Furthermore, we evaluate the net energy gain considering the Landauer limit of measurement-feedback costs, confirming the physical feasibility of this quantum vacuum heat engine.

INTRODUCTION

The concept of extracting energy from vacuum fluctuations has long been a subject of theoretical interest. Recent advances in superconducting circuits and topological insulators provide a new platform to realize non-equilibrium thermodynamic cycles at the quantum level. In this Letter, we propose an 8-site topological circuit capable of harvesting zero-point energy.

THE MODEL

We consider an 8-site ring of superconducting Josephson junctions. The Hamiltonian of the system is given by:

$$\hat{H} = \sum_{i=1}^{8} \left(\frac{\hat{p}_i^2}{2L} + \frac{1}{2} C_{\text{eff}}^{-1} \hat{q}_i^2 + \frac{K}{2} \hat{a}_i^\dagger \hat{a}_i^\dagger \hat{a}_i \hat{a}_i \right) + \hat{H}_{\text{int}}(\Phi) \quad (1)$$

where $C_{\text{eff}}^{-1} < 0$ represents the negative capacitance realized by active feedback or ferroelectric phase transitions. The interaction term $\hat{H}_{\text{int}}$ incorporates the topological pumping via a time-dependent geometric phase $\Phi_i(t) = \Omega t$.

MASTER EQUATION AND STEADY STATE

To account for dissipation and thermal noise, we employ the Lindblad master equation:

$$\dot{\rho} = -\frac{i}{\hbar}[\hat{H}, \rho] + \sum_i \gamma_i \left((\bar{n}_i + 1)\mathcal{D}[\hat{a}_i]\rho + \bar{n}_i \mathcal{D}[\hat{a}_i^\dagger]\rho \right) \quad (2)$$

where $\mathcal{D}[\hat{O}]\rho = \hat{O}\rho\hat{O}^\dagger - \frac{1}{2}\{\hat{O}^\dagger\hat{O}, \rho\}$.

Solving for the steady state $\dot{\rho} = 0$, we find a stable negative-energy manifold where the output power P_{out} exceeds the dissipation when the pumping frequency Ω exceeds the critical threshold Ω_c.

THERMODYNAMIC EFFICIENCY

The net energy gain P_{net} is evaluated by subtracting the informational cost of feedback $P_{\text{info}} \geq k_B T \dot{I} \ln 2$. Our simulation indicates that for a high-Q factor circuit ($\gamma <$ 1 MHz) at $T = 10$ mK, the topological gain from the vacuum mode-shaping overcomes the Landauer limit.

CONCLUSION

The proposed 8-site meta-material acts as a quantum vacuum heat engine. This result opens a new pathway for autonomous quantum power supplies in extreme environments.

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